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Publication list

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Summary

Indikator	Oppnåelse*
Citations	125
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Journal papers

[1] N. Summers, **J. E. Bremnes**, H. Liu, H. S. Løvås, G. M. Fragoso, and G. Johnsen, "Assessing arctic under sea-ice light regimes and ice algal bio-optical properties using rov-based hyperspectral imaging," *Polar Biology*, vol. 49, pp. 1–26, 2026. DOI: 10.1007/s00300-026-03462-5.

[2] S. D. Kristensen, R. G. Maidana, I. B. Utne, and **J. E. Bremnes**, "Evaluating the effect of risk metrics for supporting operational decision-making by autonomous surface vehicles," vol. 338, p. 121 937, 2025. DOI: https://doi.org/10.1016/j.oceaneng.2025.121937.

[3] **J. E. Bremnes**, I. B. Utne, T. R. Krogstad, and A. J. Sørensen, "Holistic risk modeling and path planning for marine robotics," *IEEE Journal of Oceanic Engineering*, vol. 50, no. 1, pp. 252–275, 2024. DOI: 10.1109/JOE.2024.3432935.

[4] **J. E. Bremnes**, T. R. Fyrvik, T. R. Krogstad, and A. J. Sørensen, "Design of a switching controller for tracking AUVs with an ASV," *IEEE Transactions on Control Systems Technology*, vol. 32, no. 5, pp. 1785–1800, 2024. DOI: 10.1109/TCST.2024.3375151.

[5] A. Vasilijevic, **J. E. Bremnes**, and M. Ludvigsen, "Remote operation of marine robotic systems and next-generation multi-purpose control rooms," *Journal of Marine Science and Engineering*, vol. 11, no. 10, pp. 1–25, 2023. DOI: 10.3390/jmse11101942.

[6] R. Yang, **J. E. Bremnes**, and I. B. Utne, "Online risk modeling of autonomous marine systems: Case study of autonomous operations under sea ice," *Ocean Engineering*, vol. 281, p. 114 765, 2023. DOI: 10.1016/j.oceaneng.2023.114765.

[7] **J. E. Bremnes**, A. H. Brodtkorb, and A. J. Sørensen, "Hybrid observer concept for sensor fusion of sporadic measurements for underwater navigation," *International Journal of Control, Automation and Systems*, vol. 19, pp. 137–144, 2020. DOI: 10.1007/s12555-019-0684-2.

[8] **J. E. Bremnes**, C. A. Thieme, A. J. Sørensen, I. B. Utne, and P. Norgren, "A Bayesian approach to supervisory risk Control of AUVs applied to under-ice operations," *Marine Technology Society Journal*, vol. 54, no. 4, pp. 16–39, 2020. DOI: 10.4031/MTSJ.54.4.5.

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[9] **J. E. Bremnes** and T. R. Krogstad, "Planning in circles: Autonomous planning for circular synthetic aperture sonar surveying with AUVs," in *Proceedings of the 2025 Undersea Defence Technology Conference*, Oslo, Norway, 2025, pp. 1–6.

[10] **J. E. Bremnes**, H. Y. Rørstad, T. R. Krogstad, and Ø. Midtgaard, "Adaptive AUV planning For robust optical object inspection using in-situ sonar data," in *Proceedings of the 2024 IEEE/OES Autonomous Underwater Vehicle Symposium (AUV)*, Boston, MA, USA, 2024, pp. 1–6. DOI: 10.1109/AUV61864.2024.11030789.

[11] A. Vasilijevic, H. Gravråkmo, K. Barstein, and **J. E. Bremnes**, "Infrastructure for remote experimentation in the Trondheim Fjord," in *Proceedings of the OCEANS 2023 Conference*, Limerick, Ireland, 2023, pp. 1–6. DOI: 10.1109/OCEANSLimerick52467.2023.10244448.

[12] R. Yang, **J. E. Bremnes**, and I. Utne, "A system-theoretic approach to hazard identification of operation with multiple autonomous marine systems (AMS)," in *Proceedings of the 32nd European Safety and Reliability Conference (ESREL)*, Dublin, Ireland, 2022, pp. 1–8. DOI: 10.3850/978-981-18-5183-4_R15-01-038-cd.

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[15] **J. E. Bremnes**, P. Norgren, A. J. Sørensen, C. A. Thieme, and I. B. Utne, "Intelligent risk-based under-ice altitude control for autonomous underwater vehicles," in *Proceedings of the IEEE/MTS OCEANS 2019 Conference*, Seattle, WA, USA, 2019, pp. 1–8. DOI: 10.23919/OCEANS40490.2019.8962532.

[16] **J. E. Bremnes**, A. H. Brodtkorb, and A. J. Sørensen, "Sensor-based hybrid translational observer for underwater navigation," in *Proceedings of the of the 12th IFAC Conference on Control Applications in Marine Systems, Robotics and Vehicles (CAMS)*, Daejeon, South Korea, 2019, pp. 378–383. DOI: 10.1016/j.ifacol.2019.12.336.

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[17] **J. E. Bremnes** and T. R. Krogstad, *CsasPlanner: Decisional autonomy for circular synthetic aperture sonar surveying with AUVs*, FFI report, exempt from public access, 2025.

[18] **J. E. Bremnes** and T. R. Krogstad, *Intelligent ID: Replanning for robust optical identification with HUGIN using EM2040*, FFI report, exempt from public access, 2025.

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[19] **J. E. Bremnes**, *Safe Autonomy in Marine Robotics*. PhD thesis, Department of Marine Technology, Norwegian University of Science and Technology, Trondheim, Norway, 2024.

[20] **J. E. Bremnes**, *Towards Robust Autonomy of Underwater Vechicles in Arctic Operations*. Master thesis, Department of Marine Technology, Norwegian University of Science and Technology, Trondheim, Norway, 2019.

Miscellaneous reports and contributions

[21] M. Ludvigsen, P. Assmy, M. J. S. Adams, M. A. Arumi, T. Mo-Bjørkelund, Y. V. Bodur, **J. E. Bremnes**, S. Flo, C. Gawinski, J. Giebichtenstein, E. Halvorsen, M. K. V. Hatlebakk, S. Hess, H. Hop, P. Itkin, E. M. Jones, K. Karlsson, A. Keck, D. Kohlbach, S. G. Kohler, R. L. Krapp, P. Leopold, M. Marquardt, E. J. Molina, R. Nowicki, L. M. Olsen, G. A. Ortiz, T. Petit, T. R. de Freitas, K. Saubrekka, A. Sen, A. Steer, N. Summers, S. Thiele, H. Thorstensen, and M. Vihtakari, *Seasonal cruise Q2 2021: Cruise report*, The Nansen Legacy Report Series, 2022. DOI: 10.7557/nlrs.6689.

[22] D. Metlay, A. L. S. Clair, B. Twomey, **J. E. Bremnes**, Ø. J. Rødseth, and S. Massaiu, *Autonomous transportation technologies: Society and individuals in the loop. Report of the discussions of breakout session*, Proceedings of the First International Workshop on Autonomous Systems Safety, 2019.

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[23] L. Aarønæs, *Han vil lære undervannsroboter å frykte døden*, ffi.no, 2024. [Online]. Available: https://www.ffi.no/aktuelt/nyheter/han-vil-laere-undervannsroboter-a-frykte-doden.

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